

Samuel Crabbendam

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Education

University of Florida, Wertheim College of Engineering

May 2028

Bachelor of Science in Aerospace Engineering

GPA: 3.6

Relevant Coursework: *Design & Manufacturing, Thermodynamics, Mechanics of Materials, Dynamics*

Skills

- Proficiency in MATLAB, Fusion360, SolidWorks, Flow5, UAS systems, Microsoft Office, 3D Print, Github

Certificates

- CSWA Certified, *SolidWorks* Gainesville, FL | 2025 - Present
- Part 107 Remote Pilot License, *West Orange High School* Winter Garden, FL | 2023 - Present
- Part 61 Private Pilot License, *Orlando Flight School* Orlando, FL | 2026 - Present
- First Aid/CPR/AED Certified; Adult and Pediatric, *UF RecSports* Gainesville, FL | 2025 - Present

Experience

Payload Mechanics Engineer / Flight Dynamics Engineer | *Swamp Launch Rocketry*

Gainesville, FL | August 2024 – Present

- Designed the 2026 IREC aluminum composite CubeSat system housing two graduate research experiments.
- Facilitated experiments analyzing gene expression and protein response under launch G-forces.
- Designed and assembled the 2025 IREC aluminum and ABS chase camera deployment system.
- Simulated rocket designs and stability in varying wind conditions, resulting in a successful launch in the Spaceport America Cup.

Fixed Wing UAS Design & Build | *Independent Project*

Gainesville, FL | July 2026 – Present

- Designing a fixed-wing reconnaissance UAS featuring GPS-guided autonomous flight, independent camera control, and a 2 ft wingspan.
- Performed aerodynamics analysis through Flow5 to validate airframe configuration prior to assembly.
- Engineering a foam & PLA airframe to maximize strength-to-weight ratio.
- Creating a servo-actuated rotating hemispherical camera gimbal for increased visibility during flight.
- Managing \$500 project budget and 16 week fabrication timeline through iterative design-build-test methodology.

Wind Tunnel Research Assistant | *Undergraduate Research at the University of Florida*

Gainesville, FL | August 2026 – Present

- Training with faculty on closed loop wind tunnel startup and procedures ahead of independent aerodynamic testing.
- Authoring operations manual for wind tunnel startup and safety, for use by future student researchers.

Project Engineering Intern | *Garney Construction*

Tampa, FL | May 2026 – August 2026

- Conducted quality control inspections of structural assemblies against engineering specifications
- Interpreted engineering drawings and specifications to generate material takeoffs and verify design compliance.
- Eliminated unnecessary sitewide production delays by creating an Excel procurement system
- Authored technical reports, meeting documentation, and project updates while coordinating across engineering, management, vendor, and field construction teams.
- Verified material quality via standardized testing to confirm specifications were met before installation.